

Lecture 1 ^{→ Data} Laws: Simple formulae that map data size
for $\downarrow$ data composition / $\downarrow$ error
data repetition
data selection for adaptive scaling

remarkably linear relationship between log-data size and log-error.

Holds across domains and models

Theory understanding: similar to generalization bounds: mean estimation example

Applications: data collection/curation

#2! scaling laws for model engineering $\rightarrow$ for model

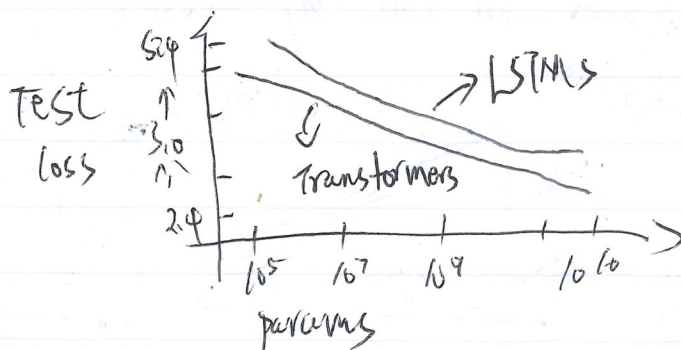
How can we efficiently design huge LMs?

#1 LSTMs VS Transformers

#2 Adam VS SGD

choices of setting the model: Architecture / optimizer / hyperparameters / batch size

1) Architecture: Transformer VS LSTM



$\Leftarrow$ spend millions of params training

LSTM | Transformer

#1) Data scaling laws: simple formulae that maps ^{data} data size
 ↓ scaling law with { data composition / mix to error
 data repetition
 data selection for adaptive scaling

↓ remarkably linear relationship between log-data size and log-error.

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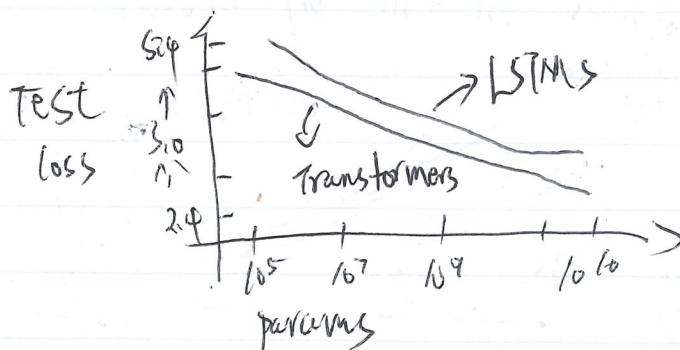
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 Batch Size

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